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年級：

科目：

練習題 10.5

$$1. (1) e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$(2) \cos^2 x = \frac{1}{2} + \cos 2x \cdot \frac{1}{2} = \frac{1}{2} + \sum_{n=0}^{\infty} \frac{(-1)^n (2x)^{2n}}{(2n)!} = \frac{1}{2} + \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} x^{2n}$$

$$\begin{aligned} (3) \frac{x^{12}}{1-x} &= -\frac{1-x^{12}}{1-x} + \frac{1}{1-x} = -\left(\sum_{n=0}^{11} x^n\right) - \sum_{n=0}^{\infty} \frac{\left(\frac{1}{x-1}\right)^{(n)} \Big|_{x=0}}{n!} x^n \\ &= -\sum_{n=0}^{11} x^n - \sum_{n=0}^{\infty} (-1)^n \cdot (-1)^{n+1} \cdot x^n = -\sum_{n=0}^{11} x^n + \sum_{n=0}^{\infty} x^n \\ &= \sum_{n=12}^{\infty} x^n \end{aligned}$$

$$\begin{aligned} (4) \ln \frac{1+x}{1-x} &= \frac{1}{2} (\ln(1+x) - \ln(1-x)) = \frac{1}{2} \left(\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n - \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} (-x)^n \right) \\ &= \frac{1}{2} \sum_{n=1}^{\infty} \frac{2}{2n-1} x^{2n-1} = \sum_{n=1}^{\infty} \frac{1}{2n-1} x^{2n-1} \end{aligned}$$

$$\begin{aligned} (5) \frac{x}{1+x-2x^2} &= \frac{x}{(1-x)(1+2x)} = \frac{1}{3} \left(\frac{1}{1-x} - \frac{1}{1+2x} \right) = \frac{1}{3} \left(\frac{1}{1-x} - \frac{1}{1-(-2x)} \right) \\ &= \frac{1}{3} \left(\sum_{n=0}^{\infty} (-2x)^n - \sum_{n=0}^{\infty} x^n \right) \\ &= \sum_{n=0}^{\infty} \left(\frac{1-(-2)^n}{3} \right) x^n \end{aligned}$$

$$(6) (1+x)e^{1+x} = (1+x) \sum_{n=0}^{\infty} \frac{1}{n!} (1+x)^n = \sum_{n=0}^{\infty} \frac{1}{n!} (1+x)^{n+1}$$

$$\begin{aligned} 2. (1) (1+x) \ln(1+x) &= (1+x) \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^{n+1} \\ &= x + \sum_{n=2}^{\infty} \frac{(-1)^{n+1}}{n} x^n + \sum_{n=2}^{\infty} \frac{(-1)^n}{n-1} x^n \\ &= x + \sum_{n=2}^{\infty} \frac{(-1)^n}{n(n-1)} x^n \end{aligned}$$

$$(2) \arctan x - \ln \sqrt{1+x^2}$$

$$\text{解. 記 } u = \arctan x, \quad u' = \frac{1}{1+x^2} \Leftrightarrow (1+x^2) u' = 1$$

$$\Rightarrow u^{(n+1)} (1+x^2) + n u^{(n)} \cdot 2x + \frac{n(n-1)}{2} u^{(n-1)} \cdot 2 = 0$$

$$\text{當 } x=0 \text{ 時, } u^{(n+1)}(0) = -n(n-1) u^{(n-1)}(0)$$

$$\text{當 } n=2k \text{ 時, } u^{(2k+1)}(0) = -2k(2k-1) u^{(2k-1)}(0) = \dots = (-1)^k (2k)! u'(0) = (-1)^k (2k)! \cdot \frac{1}{1+0} = (-1)^k (2k)!$$

$$\text{當 } n=2k-1 \text{ 時, } u^{(2k)}(0) = -(2k-1)(2k-2) u^{(2k-2)}(0) = \dots = 0$$

$$\text{所以, } \arctan x = \sum_{k=1}^{\infty} \frac{u^{(2k-1)}(0)}{(2k-1)!} x^{2k-1} = \sum_{k=1}^{\infty} \frac{(2k-2)! (-1)^{k-1}}{(2k-1)!} x^{2k-1} = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1} x^{2n-1}$$

$$\begin{aligned} \arctan x \rightarrow \ln \sqrt{1+x^2} &= \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1} x^{2n-1} - \frac{1}{2} \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^{2n} \\ &= \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1} x^{2n-1} - \frac{1}{2} \left(\sum_{n=1}^{\infty} \frac{1}{2n-1} x^{2n-1} + \sum_{n=1}^{\infty} \frac{-1}{2n} x^{2n} \right) \\ &= \sum_{n=1}^{\infty} \frac{2(-1)^{n-1}-1}{2(2n-1)} x^{2n-1} + \frac{1}{2} \sum_{n=1}^{\infty} \frac{1}{n} x^{2n} \end{aligned}$$

(3) $\frac{x}{(1-x)(1-x^2)}$

解: 設 $\frac{x}{(1-x)(1-x^2)} = \frac{A}{1-x} + \frac{B}{1+x} + \frac{C}{(1-x)^2} = \frac{A(1-x^2) + B(1-x)^2 + C(1+x)}{(1-x)(1-x^2)}$

由此可算得 $C = \frac{1}{2}$, $B = -\frac{1}{4}$, $A = -\frac{1}{4}$

$$\begin{aligned} \text{因此 } \frac{x}{(1-x)(1-x^2)} &= -\frac{1}{4} \cdot \frac{1}{1-x} + \left(-\frac{1}{4}\right) \cdot \frac{1}{1+x} + \frac{1}{2} \cdot \frac{1}{(1-x)^2} = \frac{1}{2(1-x)^2} + \frac{1}{2(1-x)^2} \\ &= \frac{1}{2} \left(\frac{1}{(1-x)^2} - \frac{1}{1-x^2} \right) = \frac{1}{2} \left(\left(\sum_{n=0}^{\infty} (-1)^n x^n \right)^2 - \sum_{n=0}^{\infty} x^{2n} \right) \\ &= \frac{1}{2} \left(\left(\sum_{n=0}^{\infty} x^n \right)^2 - \sum_{n=0}^{\infty} x^{2n} \right) \end{aligned}$$

(4) $(1+x) \arctan x = (1+x^2) \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1} x^{2n-1} = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1} x^{2n-1} + \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1} x^{2n+1}$

解: $= x + \sum_{n=2}^{\infty} \frac{(-1)^{n-1}}{2n-1} x^{2n-1} + \sum_{n=2}^{\infty} \frac{(-1)^n}{2n-3} x^{2n-1}$

$$= x + \sum_{n=2}^{\infty} \frac{(-1)^n \cdot 2}{(2n-1)(2n-3)} x^{2n-1}$$

3. (1) $\arcsin x$

解: 設 $u = \arcsin x$, $u' = \frac{1}{\sqrt{1-x^2}} \Leftrightarrow (1-x^2)(u')^2 = 1$

$$\Rightarrow [(u')^2]^{(n)} (1-x^2) + n [(u')^2]^{(n-1)} (-2x) + C_n [(u')^2]^{(n-2)} (-2) = 0.$$

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3. (1) $\arcsin x$

解. $u = \arcsin x \Rightarrow u' = \frac{1}{\sqrt{1-x^2}} \Leftrightarrow (1-x^2)(u')^2 = 1 \Rightarrow (-2x)(u')^2 + (1-x^2) \cdot 2u'u'' = 0$

因為 $u' \neq 0$, 所以有 $2(1-x^2)u'' - 2xu' = 0 \Leftrightarrow (1-x^2)u'' - xu' = 0$

$\Rightarrow u^{(n+2)}(1-x^2) + n u^{(n+1)}(-2x) + \frac{n(n-1)}{2} u^{(n)}(-2) - u^{(n+1)}x - n u^{(n)} = 0$

~~$\Leftrightarrow u^{(n+2)}(1-x^2) + u^{(n+1)}x(-2n) + u^{(n)}(1+n-n^2) = 0$~~

$\Leftrightarrow u^{(n+2)}(1-x^2) - u^{(n+1)}x(2n+1) - u^{(n)}n^2 = 0$

當 $x=0$ 時, $u^{(n+2)}(0) = n^2 u^{(n)}(0)$

當 $n=2k$ 時, $u^{(2k)}(0) = (2k-2)^2 u^{(2k-2)}(0) = \dots = [(2k-2)!!]^2 u^{(2)}(0) = 0$

當 $n=2k-1$ 時, $u^{(2k-1)}(0) = (2k-3)^2 u^{(2k-3)}(0) = \dots = [(2k-1)!!]^2 u'(0) = [(2k-1)!!]^2$

此外, $u^{(0)} = 0$

所以, $\arcsin x = \sum_{n=1}^{\infty} \frac{[(2n-1)!!]^2}{(2n-1)!} x^{2n-1} = \sum_{n=1}^{\infty} \frac{(2n-1)!!}{(2n-2)!!} x^{2n-1}$

(2) $\ln(x + \sqrt{1+x^2})$

解. $u = \ln(x + \sqrt{1+x^2}) \Leftrightarrow e^u = x + \sqrt{1+x^2} \Leftrightarrow (e^u)^2 - 2xe^u + x^2 = 1+x^2$

$\Leftrightarrow (e^u)^2 - 2xe^u = 1 \Rightarrow 2e^u \cdot e^u \cdot u' - 2(e^u + x \cdot e^u \cdot u') = 0$

$\Leftrightarrow e^u \cdot u' - (1+xu') = 0 \Leftrightarrow (e^u - x)u' = 1 \Leftrightarrow \sqrt{1+x^2} u' = 1$

$\Leftrightarrow (1+x^2)(u')^2 = 1 \Rightarrow 2u'u''(1+x^2) + (u')^2 \cdot 2x = 0$

$\Leftrightarrow u''(1+x^2) + xu' = 0$

$\Rightarrow u^{(n+2)}(1+x^2) + n u^{(n+1)}(2x) + \frac{n(n-1)}{2} u^{(n)} \cdot 2 + x u^{(n+1)} + n u^{(n)} = 0$

當 $x=0$ 時, $u^{(n+2)}(0) = -n^2 u^{(n)}(0)$

當 $n=2k$ 時, $u^{(2k)}(0) = -(2k-2)^2 u^{(2k-2)}(0) = \dots = 0$

當 $n=2k-1$ 時, $u^{(2k-1)}(0) = -(2k-3)^2 u^{(2k-3)}(0) = (-1)^{k-1} [(2k-3)!!]^2 u'(0) = (-1)^{k-1} [(2k-3)!!]^2$

因此, $\ln(x + \sqrt{1+x^2}) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} [(2n-3)!!]^2}{(2n-1)!} x^{2n-1} = x + \sum_{n=2}^{\infty} \frac{(-1)^{n-1} [(2n-3)!!]^2}{(2n-1)!} x^{2n-1}$

$= x - \sum_{n=2}^{\infty} \frac{(-1)^n (2n-1)!!}{(2n-1)^2 (2n-2)!!} x^{2n-1} = x - \sum_{n=2}^{\infty} \frac{(-1)^n (2n-3)!!}{(2n-1)(2n-2)!!} x^{2n-1}$

$$(3) \int_0^x \frac{\sin t}{t} dt$$

解: $u(x) = \int_0^x \frac{\sin t}{t} dt \Rightarrow u' = \frac{\sin x}{x} \Rightarrow xu' = \sin x \Rightarrow (xu')^{(n)} = (\sin x)^{(n)}$

當 $n=4k+1$ 時, $u^{(4k+2)} x + u^{(4k+1)} = \cos x \Rightarrow u^{(4k+1)}(0) = 1$

當 $n=4k+2$ 時, $u^{(4k+3)} x + u^{(4k+2)} = -\sin x \Rightarrow u^{(4k+2)}(0) = 0$

當 $n=4k+3$ 時, $u^{(4k+4)} x + u^{(4k+3)} = -\cos x \Rightarrow u^{(4k+3)}(0) = -1$

當 $n=4k+4$ 時, $u^{(4k+5)} x + u^{(4k+4)} = \sin x \Rightarrow u^{(4k+4)}(0) = 0$

特別地, $u^{(0)}(0) = 0, u^{(1)}(0) = 1$

因此, $\int_0^x \frac{\sin t}{t} dt = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{(2n-1)!} x^{2n-1}$

4. 設 $f(x) = Q\left(\frac{x-1}{x+1}\right) \Leftrightarrow Q(t) = f\left(\frac{1+t}{1-t}\right) \Leftrightarrow Q(x) = f\left(\frac{1+x}{1-x}\right) = \ln(1+x) - \ln(1-x)$

$\Rightarrow Q(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} x^n - \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} (-x)^n = \sum_{n=1}^{\infty} \frac{2}{2n-1} x^{2n-1}$

$\Rightarrow f(x) = \ln x = \sum_{n=1}^{\infty} \frac{2}{2n-1} \left(\frac{x-1}{x+1}\right)^{2n-1}$

5. 設 $f(x) = Q\left(\frac{x}{1+x}\right) \Leftrightarrow Q(t) = f\left(\frac{t}{1-t}\right) \Leftrightarrow Q(x) = f\left(\frac{x}{1-x}\right)$

$Q(x) = f\left(\frac{x}{1-x}\right) = \frac{\frac{x}{1-x}}{\sqrt{1+\frac{x}{1-x}}} = \frac{\frac{x}{1-x}}{1} = \frac{x}{\sqrt{1-x}} = x(1-x)^{-\frac{1}{2}}$

$= x \sum_{n=0}^{\infty} \frac{(2n+1)!!}{n!} \left(-\frac{1}{2}\right)^n x^n = \sum_{n=0}^{\infty} \frac{(2n-1)!!}{n!} \left(-\frac{1}{2}\right)^n x^{n+1} = \sum_{n=1}^{\infty} \frac{(2n-3)!!}{(n-1)!} \left(-\frac{1}{2}\right)^{n-1} x^n$

$\Rightarrow f(x) = Q\left(\frac{x}{1+x}\right) = \sum_{n=1}^{\infty} \frac{(2n-3)!!}{(n-1)!} \left(-\frac{1}{2}\right)^{n-1} \left(\frac{x}{1+x}\right)^n$

問題 10.5

(1) $f(x) = \sum_{n=0}^{\infty} \frac{\sin(2^n x)}{n!}$

因為 $\sum_{n=0}^{\infty} \left| \frac{\sin(2^n x)}{n!} \right|^{(k)} = \sum_{n=0}^{\infty} \frac{2^{kn}}{n!} \sin\left(2^n x + \frac{kn}{2}\right) \leq \sum_{n=0}^{\infty} \frac{2^{kn}}{n!}, 0 \leq \lim_{n \rightarrow \infty} \frac{2^{kn}}{\sqrt[n]{n!}} = \lim_{n \rightarrow \infty} \frac{2^k}{\sqrt[n]{n!}} \leq \lim_{n \rightarrow \infty} \frac{2^k}{\sqrt[n]{n}} = 0$

所以 $f(x)$ 是任意階連續導數, 且任意階導數一致收斂
每項均有

培道中學校

姓名：

(號)

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年級：

科目：

所以 $f(x)$ 任意階連續可導,

$$\begin{aligned} f(x) &= \sum_{n=0}^{\infty} \frac{\sin(2^n x)}{n!} \sim \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = \sum_{n=0}^{\infty} \left(2^{n^2} \sin\left(\frac{2^n \pi}{2}\right) \right) \frac{x^n}{n!} \\ &= \sum_{n=0}^{\infty} \left(\left(\sum_{k=0}^{\infty} \frac{2^{n^2}}{k!} \sin\left(\frac{k\pi}{2}\right) \right) \cdot \frac{1}{n!} x^n \right) = \sum_{n=0}^{\infty} \left(\left(\sum_{k=0}^{\infty} \frac{(-1)^n \cdot 2^{k(2n+1)}}{k!} \right) \cdot \frac{1}{(2n+1)!} x^{2n+1} \right) \\ &= \sum_{n=0}^{\infty} \left(\frac{(-1)^n \cdot 2^{(2n+1)R}}{R} \right) = \sum_{n=0}^{\infty} \left(\frac{(-1)^n}{(2n+1)!} \left(\sum_{k=0}^{\infty} \frac{(2^{2n+1})^k}{k!} \right) x^{2n+1} \right) \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n e^{2^{2n+1}}}{(2n+1)!} x^{2n+1} \end{aligned}$$

$$\text{收斂半徑 } R = \lim_{n \rightarrow \infty} \sqrt[n]{\frac{(2n+1)!}{e^{2^{2n+1}}}} \leq \lim_{n \rightarrow \infty} \frac{(2n+3)2n}{e^{\frac{2^{2n+1}}{2}}} \leq \lim_{n \rightarrow \infty} \frac{(2n+3)n^2}{2^{2n+1}} \leq \lim_{n \rightarrow \infty} \frac{(2n+3)n^2}{4^{2n+1}} = 0.$$

即 $R=0$.

因此, 在 $x=0$ 處的 Taylor 級數除 $x=0$ 外處處發散.

練習題 12.1

$$1. a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} T_n(x) dx = -$$

$$1. a_m = \frac{1}{\pi} \int_{-\pi}^{\pi} T_n(x) \cos mx dx = \frac{1}{\pi} \sum_{k=0}^n \int_{-\pi}^{\pi} (\alpha_k \cos kx \cos mx + \beta_k \sin kx \cos mx) dx$$

$$= \begin{cases} 0, & m > n \\ \frac{1}{\pi} \int_{-\pi}^{\pi} \alpha_m \cos^2 mx dx, & m \leq n \end{cases}$$

$$= \begin{cases} 0, & m > n \\ \frac{\alpha_m}{\pi} \int_{-\pi}^{\pi} \frac{1 + \cos 2mx}{2} dx, & m \leq n \end{cases}$$

$$= \begin{cases} 0, & m > n \\ \alpha_0, & m=0 \\ \alpha_m, & 0 < m \leq n \end{cases} = \begin{cases} 0, & m > n \\ \alpha_m, & 0 \leq m \leq n \end{cases}$$

$$b_m = \frac{1}{\pi} \int_{-\pi}^{\pi} (\alpha_k \cos kx \sin mx + \beta_k \sin kx \sin mx) dx$$

$$= \begin{cases} 0, & m > n \\ \frac{\beta_m}{\pi} \int_{-\pi}^{\pi} \frac{1 - \cos 2mx}{2} dx, & 0 \leq m \leq n \end{cases} = \begin{cases} 0, & m > n \\ \beta_m, & 0 \leq m \leq n \end{cases} = \begin{cases} 0, & m > n \\ \beta_m, & 0 \leq m \leq n \end{cases}$$

所以, $\forall k \in \mathbb{Z}^+$, 有 $a_k = \alpha_k$, $b_k = \beta_k$

$$\begin{aligned} 2. (1) \quad a_{2n-1} &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(2n-1)x dx = \frac{1}{\pi} \left(\int_{-\pi}^0 f(x) \cos(2n-1)x dx + \int_0^{\pi} f(x+\pi) \cos(2n-1)(x+\pi) d(x+\pi) \right) \\ &= \frac{1}{\pi} \int_{-\pi}^0 f(x) (\cos(2n-1)x - \cos(2n-1)x) dx = 0 \end{aligned}$$

$$\begin{aligned} b_{2n-1} &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(2n-1)x dx = \frac{1}{\pi} \left(\int_{-\pi}^0 f(x) \sin(2n-1)x dx + \int_0^{\pi} f(x+\pi) \sin(2n-1)(x+\pi) d(x+\pi) \right) \\ &= \frac{1}{\pi} \left(\int_{-\pi}^0 f(x) (\sin(2n-1)x - \sin(2n-1)x) dx \right) = 0. \end{aligned}$$

$$\begin{aligned} (2) \quad a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos 2nx dx = \frac{1}{\pi} \left(\int_{-\pi}^0 f(x) \cos 2nx dx + \int_0^{\pi} f(x+\pi) \cos 2n(x+\pi) d(x+\pi) \right) \\ &= \frac{1}{\pi} \int_{-\pi}^0 (f(x) - f(x)) \cos 2nx dx = 0 \end{aligned}$$

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin 2nx dx = \frac{1}{\pi} \left(\int_{-\pi}^0 f(x) \sin 2nx dx + \int_0^{\pi} f(x+\pi) \sin 2n(x+\pi) d(x+\pi) \right) \\ &= \frac{1}{\pi} \int_{-\pi}^0 (f(x) - f(x)) \sin 2nx dx = 0. \end{aligned}$$

$$\begin{aligned} 3. \quad \bar{a}_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x+h) \cos nx dx = \frac{1}{\pi} \int_{-\pi+h}^{\pi+h} f(x) \cos n(x-h) dx = \frac{1}{\pi} \left(\cos nh \int_{-\pi+h}^{\pi+h} f(x) \cos nx dx + \sin nh \int_{-\pi+h}^{\pi+h} f(x) \sin nx dx \right) \\ &= \frac{1}{\pi} \bar{a}_n \cos nh + b_n \sin nh \end{aligned}$$

$$\begin{aligned} \bar{b}_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x+h) \sin nx dx = \frac{1}{\pi} \int_{-\pi+h}^{\pi+h} f(x) \sin n(x-h) dx = \frac{1}{\pi} \left(\cos nh \int_{-\pi+h}^{\pi+h} f(x) \sin nx dx - \sin nh \int_{-\pi+h}^{\pi+h} f(x) \cos nx dx \right) \\ &= b_n \cos nh - \bar{a}_n \sin nh \end{aligned}$$

4. 因為 $a_0 \leq |a_0|$, $|a_n \cos nx + b_n \sin nx| \leq |a_n| + |b_n|$
 所以根據 Weierstrass 定理知, $\forall x \in (-\infty, +\infty)$, $\frac{a_0}{2} + \sum_{k=1}^{\infty} (a_k \cos kx + b_k \sin kx)$ 都收斂且是一致收斂。

$$\text{記 } f(x) = \frac{a_0}{2} + \sum_{k=1}^{\infty} (a_k \cos kx + b_k \sin kx)$$

考慮 $f(x)$ 的 Fourier 級數

$$\begin{aligned} \int_{-\pi}^{\pi} f(x) \cos nx \, dx &= \int_{-\pi}^{\pi} \left(\frac{a_0}{2} + \sum_{k=1}^{\infty} (a_k \cos kx + b_k \sin kx) \right) \cos nx \, dx \\ &= \int_{-\pi}^{\pi} \frac{a_0}{2} \cos nx \, dx + \int_{-\pi}^{\pi} \sum_{k=1}^{\infty} (a_k \cos kx + b_k \sin kx) \cos nx \, dx \end{aligned}$$

$$= \begin{cases} \int_{-\pi}^{\pi} \pi a_0, & n=0. \\ \int_{-\pi}^{\pi} a_n \cdot \frac{1+\cos 2nx}{2} \, dx, & n>0 \end{cases} = \begin{cases} \pi a_0, & n=0 \\ \pi a_n + \frac{a_n}{4n} \sin 2nx \Big|_{-\pi}^{\pi}, & n>0 \end{cases} = \pi a_n$$

$$\Rightarrow a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx$$

$$\text{同理可得 } b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx$$

$$f(x+2\pi) = \frac{a_0}{2} + \sum_{k=1}^{\infty} (a_k \cos k(x+2\pi) + b_k \sin k(x+2\pi)) = \frac{a_0}{2} + \sum_{k=1}^{\infty} (a_k \cos kx + b_k \sin kx) = f(x)$$

證畢。

$$\lim_{\lambda \rightarrow \infty} \int_0^1 \log x \cos^2 \lambda x dx = \lim_{\lambda \rightarrow \infty} \int_0^1 \log x \cdot \frac{1 + \cos 2\lambda x}{2} dx = \frac{1}{2} \lim_{\lambda \rightarrow \infty} \left(\int_0^1 \log x dx + \int_0^1 \log x \cos 2\lambda x dx \right)$$

$$\int_0^1 \log x dx = [x \log x - x]_0^1 = -1 - \lim_{x \rightarrow 0^+} x(\log x - 1) = -1 - \lim_{x \rightarrow 0^+} \frac{\log x}{\frac{1}{x}}$$

$$= -1 + \lim_{x \rightarrow 0^+} \frac{\log \frac{1}{x}}{\frac{1}{x}} = -1 + \lim_{t \rightarrow +\infty} \frac{\log t}{t} = -1$$

根據 Riemann-Lebesgue 定理, $\lim_{\lambda \rightarrow \infty} \int_0^1 \log x \cos 2\lambda x dx = 0$.

$$\text{所以 } \lim_{\lambda \rightarrow \infty} \int_0^1 \log x \cos^2 \lambda x dx = \frac{1}{2} \cdot (-1) = -\frac{1}{2}$$

練習題 12.2

$$1. \text{ 原式 } a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} \operatorname{sgn}(x) \cos nx dx = \frac{1}{\pi} \cdot 0$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} \operatorname{sgn}(x) \sin nx dx = \frac{2}{\pi} \int_0^{\pi} \sin nx dx = \begin{cases} 0, & n=0 \\ -\frac{2}{n\pi} [(-1)^n - 1] = \frac{2}{n\pi} (1 - (-1)^n), & n>0 \end{cases}$$

$$\text{所以 } f(x) \sim \sum_{n=1}^{\infty} \frac{4}{(2n-1)\pi} \sin(2n-1)x$$

$$\text{因為 } f(x) \text{ 滿足分段單調, 所以 } \forall x \in (-\pi, \pi), f(x) = \sum_{n=1}^{\infty} \frac{4}{(2n-1)\pi} \sin(2n-1)x$$

$$\text{取 } x = \frac{\pi}{2}, \quad 1 = \operatorname{sgn}\left(\frac{\pi}{2}\right) = \sum_{n=1}^{\infty} \frac{4}{(2n-1)\pi} \cdot (-1)^{n-1}$$

$$\Leftrightarrow \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1} = \frac{\pi}{4}$$

2. (2) (只考慮 $a \notin \mathbb{Z}$)

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} \sin ax \cos nx dx = 0, \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} \sin ax \sin nx dx$$

$$= \frac{2}{\pi} \int_0^{\pi} \sin ax \sin nx dx = \frac{1}{\pi} \int_0^{\pi} (\cos(a-n)x - \cos(a+n)x) dx$$

$$= \frac{1}{\pi} \left(\frac{1}{a-n} \sin(a-n)\pi - \frac{1}{a+n} \sin(a+n)\pi \right)$$

$$= \frac{(-1)^n \sin a\pi}{\pi} \left(\frac{1}{a-n} - \frac{1}{a+n} \right) = \frac{2n(-1)^n \sin a\pi}{\pi(a^2 - n^2)}$$

$$\text{所以 } \sin ax = \sum_{n=1}^{\infty} \frac{2n(-1)^n \sin a\pi}{\pi(a^2 - n^2)} \sin nx$$

培道中學校

姓名：

(號)

成績

年級：

科目：

$$3. a_n = \frac{2}{1} \int_0^1 (x - [x]) \cos(2\pi n x) dx = 2 \int_0^1 x \cos(2\pi n x) dx$$

$$= \cancel{x \sin} \frac{1}{n\pi} \left(x \sin(2\pi n x) \right) \Big|_0^1 - \int_0^1 \sin 2\pi n x dx = \frac{1}{2(n\pi)^2} \cos(2\pi n x) \Big|_0^1 = 0$$

$$b_n = \frac{2}{1} \int_0^1 (x - [x]) \sin(2\pi n x) dx = 2 \int_0^1 x \sin(2\pi n x) dx = \frac{1}{n\pi} \left([-x \cos 2\pi n x]_0^1 - \int_0^1 -\cos 2\pi n x dx \right)$$

$$= \frac{1}{n\pi} \left(-1 + \frac{1}{2\pi n} \sin 2\pi n x \Big|_0^1 \right) = -\frac{1}{n\pi}$$

$$a_0 = \frac{2}{1} \int_0^1 (x - [x]) dx = x^2 \Big|_0^1 = 1$$

$$\text{所以 } f(x) \sim \frac{1}{2} + \sum_{n=1}^{\infty} \left(-\frac{1}{n\pi} \right) \sin 2\pi n x = \frac{1}{2} - \sum_{n=1}^{\infty} \frac{1}{n\pi} \sin 2\pi n x$$

$$\text{當 } x \in (0, 1) \text{ 時, } f(x) = \frac{1}{2} - \sum_{n=1}^{\infty} \frac{1}{n\pi} \sin 2\pi n x$$

$$4. (2) f(x) = \begin{cases} 0, & x < 0 \\ 2x, & x \geq 0 \end{cases}$$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos\left(\frac{n\pi}{l} x\right) dx = \frac{1}{l} \int_0^l 2x \cos\left(\frac{n\pi}{l} x\right) dx$$

$$= \frac{2}{n\pi} \left(x \sin\left(\frac{n\pi}{l} x\right) \Big|_0^l - \int_0^l \sin\left(\frac{n\pi}{l} x\right) dx \right) = \frac{2}{n\pi} \cdot \frac{l}{n\pi} \cos\left(\frac{n\pi}{l} x\right) \Big|_0^l$$

$$= \frac{2l((-1)^n - 1)}{(n\pi)^2}$$

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin\left(\frac{n\pi}{l} x\right) dx = \frac{1}{l} \int_0^l 2x \sin\left(\frac{n\pi}{l} x\right) dx = \frac{2}{n\pi} \left(-x \cos\left(\frac{n\pi}{l} x\right) \Big|_0^l + \int_0^l \cos\left(\frac{n\pi}{l} x\right) dx \right)$$

$$= \frac{2}{n\pi} \left(-l \cdot (-1)^n + \frac{l}{n\pi} \sin\left(\frac{n\pi}{l} x\right) \Big|_0^l \right) = \frac{(-1)^{n+1} \cdot 2l}{n\pi}$$

$$a_0 = \frac{1}{l} \int_{-l}^l f(x) dx = \frac{1}{l} \int_0^l 2x dx = \frac{1}{l} x^2 \Big|_0^l = l$$

$$\text{所以 } f(x) = l + \sum_{n=1}^{\infty} \left(\frac{2l((-1)^n - 1)}{(n\pi)^2} \cos\left(\frac{n\pi}{l} x\right) + \frac{(-1)^{n+1} \cdot 2l}{n\pi} \sin\left(\frac{n\pi}{l} x\right) \right)$$

$$\begin{aligned}
 5. \quad a_n &= \frac{2}{3} \int_0^3 f(x) \cos\left(\frac{2\pi n}{3}x\right) dx = \frac{2}{3} \left(\int_0^1 x \cos\left(\frac{2\pi n}{3}x\right) dx + \int_1^2 \cos\left(\frac{2\pi n}{3}x\right) dx + \int_2^3 (3-x) \cos\left(\frac{2\pi n}{3}x\right) dx \right) \\
 &= \frac{1}{n\pi} \left(\left[x \sin\left(\frac{2\pi n}{3}x\right) \right]_0^1 - \int_0^1 \sin\left(\frac{2\pi n}{3}x\right) dx + \sin\left(\frac{2\pi n}{3}x\right) \Big|_1^2 + \frac{2}{3} \int_1^2 \left(\cos\left(\frac{2\pi n}{3}(3-t)\right) \right) d(3-t) \right) \\
 &= \frac{1}{n\pi} \left(\sin\left(\frac{2\pi n}{3}\right) + \frac{3}{2\pi n} \cos\left(\frac{2\pi n}{3}\right) \Big|_0^1 + \sin\left(\frac{4\pi n}{3}\right) - \sin\left(\frac{2\pi n}{3}\right) - t \sin\left(\frac{2\pi n}{3}t\right) \Big|_1^0 + \int_1^0 \sin\left(\frac{2\pi n}{3}t\right) dt \right) \\
 &= \frac{1}{n\pi} \left(\frac{3}{2\pi n} \left(\cos\left(\frac{2\pi n}{3}\right) - 1 \right) + \sin\left(\frac{4\pi n}{3}\right) + \sin\left(\frac{2\pi n}{3}\right) - \frac{3}{2\pi n} \cos\left(\frac{2\pi n}{3}t\right) \Big|_1^0 \right) \\
 &= \frac{1}{n\pi} \left(\frac{3}{2\pi n} \left(\cos\left(\frac{2\pi n}{3}\right) - 1 \right) + \frac{3}{2\pi n} \left(\cos\left(\frac{2\pi n}{3}\right) - 1 \right) \right) \\
 &= \frac{3}{(n\pi)^2} \left(\cos\left(\frac{2\pi n}{3}\right) - 1 \right)
 \end{aligned}$$

$$\begin{aligned}
 b_n &= \frac{2}{3} \int_0^3 f(x) \sin\left(\frac{2\pi n}{3}x\right) dx = \frac{2}{3} \left(\int_0^1 x \sin\left(\frac{2\pi n}{3}x\right) dx + \int_1^2 \sin\left(\frac{2\pi n}{3}x\right) dx + \int_2^3 (3-x) \sin\left(\frac{2\pi n}{3}x\right) dx \right) \\
 &= \frac{2}{3} \left(\int_0^1 x \sin\left(\frac{2\pi n}{3}x\right) dx + \int_1^2 \sin\left(\frac{2\pi n}{3}x\right) dx + \int_1^0 x \sin\left(\frac{2\pi n}{3}x\right) dx \right) \\
 &= \frac{2}{3} \int_1^2 \sin\left(\frac{2\pi n}{3}x\right) dx = -\frac{1}{n\pi} \left(-\cos\left(\frac{2\pi n}{3}x\right) \right) \Big|_1^2 = 0.
 \end{aligned}$$

$$\begin{aligned}
 a_0 &= \frac{2}{3} \int_0^3 f(x) dx = \frac{2}{3} \left(\int_0^1 x dx + \int_1^2 dx + \int_2^3 (3-x) dx \right) \\
 &= \frac{2}{3} \left(\frac{1}{2} + 1 - \frac{1}{2}(0-1) \right) = \frac{2}{3} \times 2 = \frac{4}{3}
 \end{aligned}$$

$$f(x) = \frac{2}{3} + \sum_{n=1}^{\infty} \frac{3}{(n\pi)^2} \left(\cos\left(\frac{2\pi n}{3}\right) - 1 \right) \cos\left(\frac{2\pi n}{3}x\right)$$

$$6. \quad a_0 = \lim_{k \rightarrow \infty} \frac{1}{2k\pi} \int_{-2k\pi}^{2k\pi} |\cos x| dx = \lim_{k \rightarrow \infty} \frac{1}{k\pi} \sum_{n=1}^k \int_{2(n-1)\pi}^{2n\pi} |\cos x| dx = \frac{4}{\pi}$$

$$\begin{aligned}
 a_n &= \lim_{k \rightarrow \infty} \frac{1}{2k\pi} \int_{-2k\pi}^{2k\pi} |\cos x| \cos nx dx = \lim_{k \rightarrow \infty} \frac{1}{k\pi} \int_0^{2k\pi} |\cos x| \cos nx dx = \lim_{k \rightarrow \infty} \frac{1}{k\pi} \sum_{n=1}^k \int_{2(n-1)\pi}^{2n\pi} |\cos x| \cos nx dx \\
 &= \lim_{k \rightarrow \infty} \frac{k}{k\pi} \int_0^{2\pi} |\cos x| \cos nx dx = \frac{1}{\pi} \lim_{k \rightarrow \infty} \left(\int_0^{\frac{\pi}{2}} \cos x \cos nx dx + \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} -\cos x \cos nx dx + \int_{\frac{3\pi}{2}}^{2\pi} \cos x \cos nx dx \right) \\
 &= \frac{1}{2\pi} \left(\int_0^{\frac{\pi}{2}} (\cos(n-1)x + \cos(n+1)x) dx - \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} (\cos(n-1)x + \cos(n+1)x) dx + \int_{\frac{3\pi}{2}}^{2\pi} (\cos(n-1)x + \cos(n+1)x) dx \right)
 \end{aligned}$$

培道中學校

姓名：

(號)

成績

年級：

科目：

當 $n \neq 1$ 時.

$$\begin{aligned} 2\pi a_n &= \frac{1}{n-1} \left(\sin(n-1)x \Big|_0^{\frac{\pi}{2}} - \sin(n-1)x \Big|_{\frac{\pi}{2}}^{\frac{3\pi}{2}} + \sin(n-1)x \Big|_{\frac{3\pi}{2}}^{2\pi} \right) + \frac{1}{n+1} \left(\sin(n+1)x \Big|_0^{\frac{\pi}{2}} - \sin(n+1)x \Big|_{\frac{\pi}{2}}^{\frac{3\pi}{2}} + \sin(n+1)x \Big|_{\frac{3\pi}{2}}^{2\pi} \right) \\ &= \frac{1}{n-1} \left(-\cos \frac{n\pi}{2} - \left(\cos \frac{3n\pi}{2} + \cos \frac{n\pi}{2} \right) + \left(-\cos \frac{3n\pi}{2} \right) \right) = \frac{2}{1-n} \cdot 2 \cos \frac{n\pi}{2} = \frac{4}{n-1} \cos \frac{n\pi}{2} \\ &= \frac{1}{n-1} \left(-\cos \frac{n\pi}{2} - \left(\cos \frac{3n\pi}{2} + \cos \frac{n\pi}{2} \right) - \cos \frac{3n\pi}{2} \right) + \frac{1}{n+1} \left(\cos \frac{n\pi}{2} - \left(-\cos \frac{3n\pi}{2} - \cos \frac{n\pi}{2} \right) + \cos \frac{3n\pi}{2} \right) \\ &= \left(\cos \frac{n\pi}{2} + \cos \frac{3n\pi}{2} \right) \left(\frac{2}{n+1} - \frac{2}{n-1} \right) = 2 \cos \frac{n\pi}{2} \cdot \frac{-4}{n^2-1} = \frac{8}{1-n^2} \cos \frac{n\pi}{2} \end{aligned}$$

當 $n=1$ 時.

$$\begin{aligned} 2\pi a_n &= \left(x \Big|_0^{\frac{\pi}{2}} - x \Big|_{\frac{\pi}{2}}^{\frac{3\pi}{2}} + x \Big|_{\frac{3\pi}{2}}^{2\pi} \right) + \frac{1}{n+1} \left(\cos \frac{n\pi}{2} - \left(-\cos \frac{3n\pi}{2} - \cos \frac{n\pi}{2} \right) + \cos \frac{3n\pi}{2} \right) \\ &= \frac{2}{n+1} \cdot 2 \cos \frac{n\pi}{2} = \frac{4}{n+1} \cos \frac{n\pi}{2} = 2 \cos \frac{\pi}{2} = 0. \end{aligned}$$

$$b_n = \lim_{k \rightarrow \infty} \frac{1}{2k\pi} \int_{-2k\pi}^{2k\pi} |\cos x| \sin nx \, dx = \lim_{k \rightarrow \infty} 0 = 0.$$

所以, $|\cos x| = \frac{2}{\pi} + \frac{1}{2\pi} \sum_{n=2}^{\infty} \frac{8}{1-n^2} \cos \frac{n\pi}{2} \cos nx$

$$\begin{aligned} &= \frac{2}{\pi} + \frac{4}{\pi} \sum_{k=1}^{\infty} \frac{(-1)^k}{1-(2k)^2} \cos 2kx \\ &= \frac{2}{\pi} + \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{4n^2-1} \cos 2nx \quad (\text{滿足分段可微條件}) \end{aligned}$$

對於 $|\sin x|$,

$$a_0 = \lim_{k \rightarrow \infty} \frac{1}{2k\pi} \int_{-2k\pi}^{2k\pi} |\sin x| \, dx = \lim_{k \rightarrow \infty} \frac{1}{k\pi} \int_0^{2k\pi} |\sin x| \, dx = \lim_{k \rightarrow \infty} \frac{2k}{k\pi} \int_0^{\pi} \sin x \, dx = \frac{4}{\pi}$$

$$a_n = \lim_{k \rightarrow \infty} \frac{1}{2k\pi} \int_{-2k\pi}^{2k\pi} |\sin x| \cos nx \, dx = \lim_{k \rightarrow \infty} \frac{k}{k\pi} \int_0^{2k\pi} |\sin x| \cos nx \, dx = \lim_{k \rightarrow \infty} \frac{k}{k\pi} \int_0^{2k\pi} \sin x \cos nx \, dx$$

$$= \frac{1}{\pi} \left(\int_0^{\pi} \sin x \cos nx \, dx + \int_{\pi}^{2\pi} (-\sin x) \cos nx \, dx \right) = \frac{1}{\pi} \int_0^{\pi} \sin x \cos nx (1+(-1)^n) \, dx$$

當 n 是奇數時, $a_n = 0$

當

$$= \frac{1}{\pi} \left(-\frac{1}{n+1} \cos(n+1)x \Big|_0^\pi + \frac{1}{n-1} \cos(n-1)x \Big|_0^\pi \right) = \frac{1}{\pi} \left(\frac{1}{n-1} (-1-1) - \frac{1}{n+1} (-1-1) \right)$$

$$= -\frac{2}{\pi} \cdot \frac{2}{n^2-1} = \frac{4}{\pi} \cdot \frac{1}{1-n^2}$$

$$b_n = \lim_{k \rightarrow +\infty} \frac{1}{2k\pi} \int_{-2k\pi}^{2k\pi} |\sin x| \sin nx \, dx = 0$$

因此, $|\sin x| = \frac{2}{\pi} + \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{1-(2n)^2} \cos 2nx = \frac{2}{\pi} - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{4n^2-1} \cos 2nx$ (满足分段^{可微}条件)

$$7. \quad a_n = \frac{1}{\pi} \int_0^{2\pi} e^{ax} \cos nx \, dx = \frac{1}{n\pi} \left(e^{ax} \sin nx \Big|_0^{2\pi} - \int_0^{2\pi} a \sin nx \cdot e^{ax} \, dx \right) = \frac{a}{n\pi} \left(e^{ax} \cos nx \Big|_0^{2\pi} - \int_0^{2\pi} a \cos nx \cdot e^{ax} \, dx \right)$$

$$= \frac{a}{n^2\pi} (e^{2\pi a} - 1) - a \int_0^{2\pi} e^{ax} \cos nx \, dx = \frac{a}{n^2\pi} (e^{2\pi a} - 1 - \pi a_n)$$

$$\Rightarrow a_n \left(1 + \left(\frac{a}{n} \right)^2 \right) = \frac{a}{n^2\pi} (e^{2\pi a} - 1) \Rightarrow a_n = \frac{\frac{a}{n^2\pi} (e^{2\pi a} - 1) \cdot n^2}{n^2 + a^2} = \frac{a}{\pi} \cdot \frac{e^{2\pi a} - 1}{n^2 + a^2}$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} e^{ax} \sin nx \, dx =$$

$$a \int_0^{2\pi} e^{ax} \cos nx \, dx = -\frac{a}{n} b_n$$

練習題 12.4

$$6. \left| \sum_{k=1}^n a_k \right| = \left| \sum_{k=1}^n \left(\frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos kx dx \right) \right| = \frac{1}{\pi} \left| \int_{-\pi}^{\pi} f(x) \left(\sum_{k=1}^n \cos kx \right) dx \right|$$

$$= \frac{1}{\pi} \left| \int_{-\pi}^{\pi} f(x) \frac{\sin(n+\frac{1}{2})x - \sin \frac{1}{2}x}{2 \sin \frac{1}{2}x} dx \right|$$

設 $|f(x)| \leq M$, 則 $\left| \sum_{k=1}^n a_k \right| \leq \frac{M}{2\pi} \int_{-\pi}^{\pi} \left| \frac{\sin(n+\frac{1}{2})x - \sin \frac{1}{2}x}{\sin \frac{1}{2}x} \right| dx$

$$\Rightarrow \left| \sum_{k=1}^n a_k \right| \text{ 有界}$$

同理可得, $\left| \sum_{k=1}^n b_k \right| \leq \frac{M}{2\pi} \int_{-\pi}^{\pi} \left| \frac{\cos \frac{1}{2}x - \cos(n+\frac{1}{2})x}{\sin \frac{1}{2}x} \right| dx \Rightarrow \left| \sum_{k=1}^n b_k \right| \text{ 有界}$

由 Dirichlet 判別法知,

$$\sum_{n=1}^{\infty} \frac{a_n}{n}, \sum_{n=1}^{\infty} \frac{b_n}{n} \text{ 收斂}$$